

A note on Nacireman suprasegmental phonology

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Abstract

This note discusses some recent findings in Nacireman phonology. This language makes suprasegmental use of overtones on each syllable to encode lexical meaning. This typologically unusual feature is integrated into several morphological alternations in the language.

1 Introduction

Nacireman is the language of the Nacirema, a north American tribe.¹ Although anthropological aspects of Nacireman culture have been extensively studied (see Miner, 1956; Muller, 1971; Thompson, 1972; Walker, 1970), the language remains underdocumented, with the majority of prior research focusing on sociocultural and pragmatic influences on speech codes (Philipsen, 1992). This brief discussion note outlines some recent surprising findings in Nacireman suprasegmental phonology.

2 Background on Nacireman vowels

Like many languages, Nacireman has a straightforward system of four monophthongs /i, ɨ, ɜ, ɐ/ (cf. Walker, 1970). In addition, there are three diphthongs /eɨ, ɜɨ, ɐɨ/. Notably, /ɐɨ/ patterns

¹Also known by the endonym /mɜɨkɨn/.

phonotactically with the monophthongs, and tentative historical work suggests that it may derive from a long monophthong /ɜ:/ . In the following notes, I group it with the monophthongs and treat it as such.

3 Suprasegmental overtones

The most curious feature of Nacireman phonology is that it features *overtones*, whereby each syllable receives either a high or a low tone. This is a form of secondary articulation, similar to “overtone singing” (also known as “throat singing”) popular in many indigenous cultures around the world (Adams, 2013; Pegg, 1992). These overtones, however, convey phonemic contrast.²

There are several (near) minimal pairs which are distinguished only by the overtone. Following IPA conventions, we denote a high overtone as an acute over a vowel $\acute{V}$, and a low overtone as a grave $\grave{V}$:³

- (1) a. /k^hét^h/ ‘*Felis catus*’
b. /k^hè^h/ ‘portable bed’
- (2) a. /sɪ́t^h/ ‘warning, ultimatum’
b. /sɪ̀t^h/ ‘think.PAST’
- (3) a. /kɪ́ɔ̃p^h/ ‘berry of *Vitus* vine’
b. /kɪ̀ɔ̃p^h/ ‘to grasp, to fondle’
- (4) a. /sɪ́t^h/ ‘to rest the lower body while the upper body is upright’
b. /sɪ̀t^h/ ‘dark powder from a fire’
- (5) a. /ɪ́t/ ‘grasses of order *Poales*’
b. /ɪ̀t/ ‘unpleasant, ill-mannered’

²Preliminary acoustic investigation suggests that these overtones are primarily realized through manipulation of the second formant frequency (F_2), a dimension which in other languages is generally reserved for signaling the place of articulation of consonants.

³Abbreviations used in this paper: PAST, past tense; SG, singular; PL, plural; CAUS, causative.

- (6) a. /pɹɛ̃in/ ‘saltwater’
 b. /pɹèin/ ‘dark orange’

3.1 Overtones and diphthongs

Regarding the diphthongs /ɛ̃i, ɜ̃i/, the intensity of the overtone tends to increase towards the end of the sound. As such, it may be more accurate to term the overtones on diphthongs “rising” and “falling” rather than “high” and “low”, at least phonetically. (See Section 4 for evidence suggesting that the overtones on the diphthongs are phonologically the same as the overtones on the monophthongs.) An additional consideration is that the diphthong /ɜ̃i/ appears to be bound by phonotactic restrictions, such that it can only occur with a high overtone—see (7).

- (7) a. /pɜ̃i/ ‘male child’
 b. */pèi/

Because of this gap in the paradigm, it appears that /ɜ̃i/ has shifted in the phonetic space such that the beginning of the diphthong has a much lower overtone than would be expected otherwise (e.g. when compared with /ɛ̃i/).

3.2 Exhaustivity

Finally, there is one exception to the rule that each syllable must have a specified overtone; the vowel /ɜ/ is able to occur with no overtone, leaving the vowel neutral and somewhat targetless. An example is provided in (8), and a minimal triplet is exemplified in (9).

- (8) /kɜn/ ‘firearm’
- (9) a. /pɜ́tʰ/ ‘wager’
 b. /pɜtʰ/ ‘buttocks’
 c. /pɜ̀tʰ/ ‘purchase.PAST’

4 Suprasegmental morphophonology

Far from being a lexical oddity in and of itself, it would also appear that the overtone system of Nacireman bears is morphologically active. Consider the data in (10), (11), (12).

- (10) a. /fít^h/ ‘foot.SG’
 b. /fít^h/ ‘foot.PL’
- (11) a. /mèis/ ‘mouse.SG’
 b. /méis/ ‘mouse.PL’
- (12) a. /fit/ ‘edible item’
 b. /fít/ ‘eat.CAUS’

As shown in (13), there are several high-toned verbs which switch to low tone to indicate past tense.

	PRESENT	PAST	GLOSS
a.	pɪóɪk ^h	pɪòɪk ^h	‘shatter’
b.	t ^h óɪɪ	t ^h ɪòɪɪ	‘rip’
(13) c.	póɪɪ	pɪòɪɪ	‘carry’
d.	swóɪɪ	swɪòɪɪ	‘to make an oath’
e.	fèint	fèint	‘discover’
f.	kɪréint	kɪrèint	‘scrape’

However, these patterns are not necessarily productive. Consider /mìs/ ‘elk.SG’, but */mís/, which my informants (Harobed Nitram and Nai Rolyat, to whom I am indebted) emphatically rejected with laughter.⁴ Overall, these patterns are sporadic, with the majority of lexemes conforming to regular affixing morphology. It may be that these tonal patterns are remnants of a

⁴The correct plural is apparently /mìs3z/, which follows the more common suffixing method of pluralization in Nacireman.

historically more comprehensive system, now lost due to language contact.

5 Conclusion

This system of overtones raises interesting questions for our understanding of both Nacireman phonology in particular and the human language faculty in general. How ought we to represent such overtones phonologically—with features, autosegments, or something else? How are these overtones represented in the speaker’s mental lexicon? In language processing, is this suprasegmental information processed subsequent to segmental information, or in parallel? If Nacireman, like other languages, also uses F_2 to mark consonant place, how do listeners resolve F_2 conflicts between overtone specification and consonant transitions? It is hoped that these questions, and many more, will provide many decades’ worth of investigation, publications, grant applications, and debates.

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